

1 GPU-accelerated Bayesian calibration of a variational
2 multiphase model of multispecies biofilms [final title
3 after the results]

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15 **Abstract**

Biofilms are living materials whose composition, and with it their mechanical response, is set by interactions between species that cannot be measured directly. We calibrate a thermodynamically consistent multiphase model of a five-species oral biofilm, derived from the extended Hamilton principle, against in-vitro time courses under four combinations of community state and cultivation. All 15 entries of the symmetric interaction matrix are inferred jointly by transitional Markov chain Monte Carlo, with the stiff forward model compiled in JAX and evaluated for all particles in parallel on a GPU ($\sim 150\times$ faster than on a CPU). The data enter in stages, from species composition to community viability, and every stage is accepted only after explicit checks of seed agreement, mixing and posterior mass at the prior bounds. [Results: fit, which interactions are identified, the Fn–Pg coefficient, pH check, knockout prediction — after the final runs.]

16 *Keywords:* Bayesian inference, transitional Markov chain Monte Carlo,
17 extended Hamilton principle, multiphase continuum, biofilm, GPU

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1. Introduction

Biofilms are soft, living materials: microbial cells embedded in a self-produced matrix whose composition evolves in time and determines their mechanical response [1]. On dental implants, a shift in the species composition of the biofilm from a commensal towards a dysbiotic community is associated with peri-implantitis [2, 3, 4]. Which interactions between the species drive such a shift is difficult to measure: co-culture experiments report how the composition changes over time, not the interactions themselves.

Continuum mechanics offers a route from composition to interaction. Based on the extended Hamilton principle [5], Klempt et al. [6] derived a multiphase model in which species volume fractions and viabilities evolve under a free energy that contains a symmetric interaction matrix $\mathbf{A}$; thermodynamic consistency and the symmetry of $\mathbf{A}$ follow from the variational structure. Calibrating $\mathbf{A}$ against data turns the model into a tool to ask which interactions a measured composition requires. Bayesian updating has been used to calibrate constitutive parameters of biofilm models with surrogates [7], but the interaction matrix poses two specific difficulties. First, with five usable time points and five species, at most 25 observations constrain 15 unknowns, so identifiability cannot be assumed. Second, the coupled volume-fraction and viability dynamics produce posteriors that are multimodal and bounded by the prior box, for which the convergence of standard samplers is hard to verify.

Here we infer all 15 entries of $\mathbf{A}$ from the time courses of Heine et al. [8] for five oral taxa—*Streptococcus oralis* (So), *Actinomyces naeslundii* (An), *Veillonella* spp. (Vei), *Fusobacterium nucleatum* (Fn) and *Porphyromonas gingivalis* (Pg)—in four conditions: commensal or dysbiotic community, each under static or dynamic (HOBIC flow reactor) cultivation. We combine three elements: (i) transitional Markov chain Monte Carlo (TMCMC) [9, 10] with the forward model compiled in JAX [11] and evaluated for all particles in parallel on a GPU; (ii) a staged introduction of the data, from species composition to community viability; and (iii) explicit acceptance criteria for every stage, including a rule for widening prior bounds at which posterior mass accumulates. **[One sentence on the main result after the final runs.]**

52 2. Model

53 At a material point, species $i = 1, \dots, 5$ occupies the volume fraction
 54 $\phi_i \in (0, 1)$, of which the fraction $\psi_i \in (0, 1)$ is alive; the living volume fraction
 55 is $\bar{\phi}_i = \phi_i \psi_i$. A void fraction ϕ_0 closes the volume, $\sum_{l=0}^5 \phi_l = 1$. Following
 56 Klempt et al. [6], the evolution equations follow from the stationarity of the
 57 rate functional

$$\dot{\Psi} + \Delta + C \rightarrow \text{stat}_{(\bar{\phi}, \dot{\psi}, \gamma)}, \quad (1)$$

58 the local form of the extended Hamilton principle [5, 12, 13, 14], with the
 59 free energy, dissipation and constraint

$$\Psi = -\frac{1}{2} c^* \bar{\phi}^\top \mathbf{A} \bar{\phi}, \quad \Delta = \frac{1}{2} \dot{\bar{\phi}}^\top \boldsymbol{\eta} \dot{\bar{\phi}} + \frac{1}{2} \dot{\bar{\phi}}^\top \boldsymbol{\eta} \dot{\bar{\phi}}, \quad C = \gamma \left(\sum_{l=0}^5 \phi_l - 1 \right). \quad (2)$$

60 Here c^* is the nutrient concentration, $\boldsymbol{\eta} = \text{diag}(\eta_i)$ with $\eta_i = 1$, and γ
 61 is a Lagrange multiplier; the antibiotic term of the original model vanishes
 62 because the experiments contain no antibiotic. Stationarity gives $2n+2 = 12$
 63 coupled non-linear equations, for $i = 1, \dots, 5$,

$$0 = -c^* \psi_i (\mathbf{A} \bar{\phi})_i + \eta_i (\dot{\phi}_i \psi_i^2 + \bar{\phi}_i \dot{\psi}_i + \dot{\phi}_i) + \gamma, \quad (3)$$

$$0 = -c^* \phi_i (\mathbf{A} \bar{\phi})_i + \eta_i (\dot{\psi}_i \phi_i^2 + \bar{\phi}_i \dot{\phi}_i) + \gamma, \quad (4)$$

64 together with $0 = \gamma + \dot{\phi}_0$ and the volume constraint. They are integrated
 65 by an implicit Euler scheme with Newton iterations (2500 steps over the 21-
 66 day window), with a logarithmic barrier keeping ϕ_i, ψ_i inside $(0, 1)$. Because
 67 the free energy is quadratic in $\bar{\phi}$, $\mathbf{A}$ is symmetric and has 15 independent
 68 entries, which form the parameter vector $\boldsymbol{\theta} \in \mathbb{R}^{15}$. The coefficient a_{ij} is an
 69 effective, undirected interaction of the model; it lumps mechanisms that the
 70 model does not resolve, such as metabolite exchange or co-aggregation. Only
 71 the product $c^* a_{ij}$ enters, so we fix $c^* = 25$ to keep $a_{ij} = \mathcal{O}(1)$. Removing a
 72 species is represented by holding its volume fraction at zero.

73 3. Bayesian inference

74 *Likelihood.* The data are the species fractions at Days 3, 6, 10, 15 and 21
 75 (Day 1 is the initial condition), averaged over eight biological replicates [8].

76 The model fractions are renormalised over the five species, $\hat{y}_i = \phi_i / \sum_j \phi_j$,
 77 and compared through a weighted Gaussian log-likelihood

$$\ln p(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\theta}) = -\frac{1}{2} \sum_k \sum_{i=1}^5 w_{ik} \frac{(y_{\text{obs},i}(t_k) - \hat{y}_i(t_k; \boldsymbol{\theta}))^2}{\sigma_i^2} + \text{const}, \quad (5)$$

78 with σ_i the replicate standard deviation of species i . The weights up-weight
 79 Pg ($\lambda_{\text{Pg}} = 5$), whose late increase is the main dynamic feature of the dysbi-
 80 otic data, and the last two time points ($\lambda_{\text{late}} = 3$), and down-weight species
 81 below 5% (0.1); without these weights the median of the key coefficient is
 82 unchanged but its interval widens (Supplementary Section S1). In the later
 83 stages a second Gaussian channel compares the predicted community viabil-
 84 ity $\sum_i \bar{\phi}_i / \sum_i \phi_i$ with the measured fraction of membrane-intact cells. The
 85 measured pH under dynamic cultivation is not used in the inference and
 86 serves as a check (Section 4).

87 *Prior and stages..* The prior is uniform on a box $[l_k, u_k]$ for each entry, set to
 88 contain earlier estimates; it encodes no sign constraint. Inference proceeds in
 89 four stages on the same data: (i) composition likelihood on the box; (ii) as (i),
 90 on the box narrowed to the stage-(i) posterior mean ± 4 standard deviations;
 91 (iii) composition and viability, back on the full box, starting from the stage-
 92 (ii) particles; (iv) as (iii), on the box narrowed to the stage-(iii) posterior
 93 mean ± 4 standard deviations. A ± 4 SD box removes less than 10^{-4} of a
 94 Gaussian marginal, so narrowing shortens the tempering without truncating
 95 the posterior. If more than 20% of the posterior mass of an entry lies in the
 96 outer 5% of the box on one side, the bound is moved outwards and the stage
 97 is repeated; the boxes finally used are listed in Supplementary Table S1.

98 *TMCMC on the GPU..* TMCMC [9, 15] passes a particle population through
 99 the tempered posteriors $p_m(\boldsymbol{\theta}) \propto p(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\theta})^{\beta_m} p(\boldsymbol{\theta})$, $0 = \beta_0 < \dots < \beta_M = 1$,
 100 choosing each increment so that the coefficient of variation of the impor-
 101 tance weights stays at a target value [10]. After resampling, every particle
 102 performs K Metropolis–Hastings steps with random-walk proposals scaled
 103 by the weighted particle covariance, alternating with differential-evolution
 104 proposals [16] that follow strongly correlated directions; proposals outside
 105 the box are rejected without solving the model. Each likelihood evaluation
 106 requires one solution of Eqs. (3)–(4), and a run requires 10^5 – 10^6 of them. We
 107 implement the implicit solver, including Jacobian assembly and linear solve,
 108 in JAX, compile it into a single XLA kernel and evaluate all particles at once

109 with `vmap`. On one GPU (NVIDIA RTX 4090) this is 170 times faster per
 110 stage than a 12-core CPU at matched settings (147 times end to end), which
 111 makes it affordable to repeat every stage with three seeds.

112 *Acceptance criteria.* Every stage is run with three independent seeds and
 113 accepted only if (a) the stored log-likelihood of every particle agrees with a
 114 recomputation; (b) the tempering reaches $\beta = 1$; (c) each particle is moved
 115 at least twice per tempering step and at least five times per free parameter
 116 over the run; (d) the maximum log-likelihood differs by at most 1 across
 117 seeds; and (e) every posterior median differs across seeds by at most half
 118 a posterior standard deviation. Criterion (c) counts accepted moves rather
 119 than the acceptance rate: with optimally scaled random-walk proposals [17],
 120 $5d$ accepted moves displace each coordinate by about five posterior standard
 121 deviations, independently of the dimension d . Stages that fail are repeated
 122 with more particles or mutation steps, never interpreted. The evidence esti-
 123 mate of TMCMC is used only to compare models on the same data and box
 124 (Section 4).

125 4. Results and discussion

126 [After the final runs (budget 1300 + 550 words, figures 2–6): fits
 127 in four conditions (Fig. 2); inferred matrices (Fig. 3) and posteriors
 128 with identified/unidentified entries (Fig. 4); the Fn–Pg coefficient:
 129 stage (i)–(iv), Bayes factor, likelihood weights; identifiability: wide
 130 box with and without a Gaussian prior, Phase 1 vs. Phase 2 (Fig. 5);
 131 pH check with a fixed regression on species fractions and the Fn-
 132 knockout prediction (Fig. 6); limitations: unit-sum constraint, ef-
 133 fective parameters; outlook: propagation to mechanics (stiffness of
 134 the biofilm from its composition).]

135 5. Conclusions

136 [After the final runs.]

137 Declaration of competing interest

138 The authors declare no competing interests.

139 **Data and code availability**

140 Code, run configurations, convergence records and posterior samples are
141 archived at Zenodo ([\[DOI\]](#)) and developed at [https://github.com/keisuke58/](https://github.com/keisuke58/Tmcmc202601)
142 [Tmcmc202601](#). The experimental data are those of Heine et al. [8].

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